

Chapter 1

Sample mean

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Sample variance

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Sample standard deviation

$$s_x = \sqrt{s_x^2}$$

Inter-quartile range

$$IQR = Q_3 - Q_1$$

1.5IQR rule

$$Q_1 - 1.5IQR, Q_3 + 1.5IQR$$

Chapter 2

Axioms of Probability :

- (a) $P(S) = 1$.
- (b) For an event A, $0 \leq P(A) \leq 1$.
- (c) If events A and B are mutually exclusive, then $P(A \cup B) = P(A) + P(B)$.

For any event A,

$$P(A^c) = 1 - P(A).$$

For an empty set $\emptyset$,

$$P(\emptyset) = 0.$$

For any two events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A and B are mutually exclusive, then

$$P(A \cap B) = 0.$$

For any two events A and B,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}.$$

If A and B are independent, then

$$P(B|A) = P(B) \text{ and } P(A \cap B) = P(A)P(B).$$

The law of total probability

$$P(B) = P(A)P(B|A) + P(A^c)P(B|A^c)$$

Bayes rule

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(A^c)P(B|A^c)}$$

De Morgan's Law

$$(A \cup B)^c = A^c \cap B^c \text{ and } (A \cap B)^c = A^c \cup B^c$$

For a discrete r.v. X, its pmf is

$$p(x) = P(X = x).$$

$p(x)$ is non-negative and

$$\sum_{\text{all } x} p(x) = 1.$$

The mean and variance of X are

$$\mu_X = \sum_{\text{all } x} x \cdot p(x)$$

$$\sigma_X^2 = \sum_{\text{all } x} (x - \mu_X)^2 \cdot p(x) = \left[\sum_{\text{all } x} x^2 \cdot p(x) \right] - \mu_X^2.$$

For a continuous r.v. X with a pdf $f(x)$ and two real numbers a and b where $a < b$,

$$P(a < X < b) = \int_a^b f(x)dx.$$

$f(x)$ is non-negative and

$$\int_{-\infty}^{\infty} f(x)dx = 1.$$

The mean and variance of X are

$$\mu_X = \int_{-\infty}^{\infty} x \cdot f(x)dx$$

$$\sigma_X^2 = \int_{-\infty}^{\infty} (x - \mu_X)^2 \cdot f(x)dx = \left[\int_{-\infty}^{\infty} x^2 \cdot f(x)dx \right] - \mu_X^2.$$

The median x_m of X can be solved with

$$\int_{-\infty}^{x_m} f(x)dx = \frac{1}{2}.$$

Chapter 3

If $Y = X + c$, then

$$\sigma_Y = \sigma_X.$$

If $Y = cX$, then

$$\sigma_Y = |c|\sigma_X.$$

If $Y = c_1X_1 + \cdots + c_nX_n$, where X_i 's are independent measurements and each has uncertainty σ_{X_i} . Then

$$\sigma_Y = \sqrt{c_1^2\sigma_{X_1}^2 + \cdots + c_n^2\sigma_{X_n}^2}.$$

If

$$\bar{X} = \frac{X_1 + \cdots + X_n}{n},$$

where each and every X_i has the same uncertainty σ . Then

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}.$$

If $Y = u(X)$, where $u(\cdot)$ is a nonlinear function. Then

$$\sigma_Y \approx \left| \frac{du(X)}{dX} \right| \sigma_X.$$

Chapter 4

Bernoulli Distributions

$X \sim \text{Bernoulli}(p)$, then

$$p(x) = p^x(1-p)^{1-x} \text{ for } x = 0, 1$$

$$\mu_X = p \text{ and } \sigma_X^2 = p(1-p).$$

Binomial Distributions

$X \sim \text{Bin}(n, p)$, then

$$p(x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \text{ for } x = 0, 1, \dots, n$$

$$\mu_X = np \text{ and } \sigma_X^2 = np(1-p).$$

$$P(X = x) = \text{dbinom}(x, n, p)$$

$$P(X \leq x) = \text{pbinom}(x, n, p)$$

Poisson Distributions

$X \sim \text{Poisson}(\lambda)$, then

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!} \text{ for } x = 0, 1, \dots$$

$$\mu_X = \lambda \text{ and } \sigma_X^2 = \lambda.$$

$$P(X = x) = \text{dpois}(x, \lambda)$$

$$P(X \leq x) = \text{ppois}(x, \lambda)$$

Normal Distributions

$X \sim N(\mu, \sigma^2)$, then

$$\mu_X = \mu \text{ and } \sigma_X^2 = \sigma^2.$$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$$

$$P(X \leq x) = \text{pnorm}(x, \mu, \sigma)$$

If $P(X \leq x) = q$, then

$$x = \text{qnorm}(q, \mu, \sigma)$$

z-score is

$$z = \frac{x - \mu}{\sigma}$$

Sampling Distribution of $\bar{X}$

Let $X_1, \dots, X_n$ be a random sample from a population with mean μ and variance σ^2 . Let $\bar{X}$ be the sample mean. Then

$$\mu_{\bar{X}} = \mu \text{ and } \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}.$$

CLT : $\bar{X}$ follows a normal distribution approximately when n is sufficiently large.

z-score is

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Sampling Distribution of S_n

Let $X_1, \dots, X_n$ be a random sample from a population with mean μ and variance σ^2 . Let S_n be the sample total. Then

$$\mu_{S_n} = n\mu \text{ and } \sigma_{S_n}^2 = n\sigma^2.$$

CLT : S_n follows a normal distribution approximately when n is sufficiently large.

z-score is

$$z = \frac{s_n - n\mu}{\sqrt{n}\sigma}$$

Chapter 5

Under proper conditions (that is, you will need to consider sample sizes, population distributions, etc.) :

$(1 - \alpha)100\%$ confidence interval for μ when σ is known :

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

$(1 - \alpha)100\%$ confidence interval for μ when σ is unknown :

$$\bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}.$$

The multipliers can be found via R :

$$z_{\alpha/2} = \text{qnorm}(\alpha/2, \text{lower.tail} = F),$$

$$t_{n-1, \alpha/2} = \text{qt}(\alpha/2, n-1, \text{lower.tail} = F).$$

The sample size required for a $(1 - \alpha)100\%$ confidence interval for μ to have a precision of $\pm m$ is

$$n = \left(\frac{z_{\alpha/2} \sigma}{m} \right)^2.$$

If two samples are independent, $(1 - \alpha)100\%$ confidence interval for $\mu_X - \mu_Y$ when population variances are not equal :

$$(\bar{x} - \bar{y}) \pm t_{df, \alpha/2} \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}},$$

where

$$df = \frac{\left(\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y} \right)^2}{\frac{\left(\frac{s_X^2}{n_X} \right)^2}{n_X - 1} + \frac{\left(\frac{s_Y^2}{n_Y} \right)^2}{n_Y - 1}}.$$

If two samples are independent, $(1 - \alpha)100\%$ confidence interval for $\mu_X - \mu_Y$ when population variances are equal :

$$(\bar{x} - \bar{y}) \pm t_{df, \alpha/2} s_p \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}},$$

where

$$df = n_X + n_Y - 2 \text{ and}$$

$$s_p = \sqrt{\frac{(n_X - 1)s_X^2 + (n_Y - 1)s_Y^2}{n_X + n_Y - 2}}.$$

If sampled data are matched pairs, $(1 - \alpha)100\%$ confidence interval for $\mu_X - \mu_Y$ when population variances are equal :

$$\bar{D} \pm t_{n_d-1, \alpha/2} \frac{s_d}{\sqrt{n_d}}.$$

If a set of data has been given, you may use `t.test()` in R to find a CI. Please refer to R documentation for more info.

Chapter 6

Conclusion 1 :

Reject H_0

Statistical significant

$p\text{-value} \leq \alpha$

$(1 - \alpha)100\%$ CI doesn't contain μ_0

$|\text{test statistics}| \geq |\text{critical point}|$

Observed results likely not due to chance

Data not consistent with H_0

Conclusion 2 :

Fail to reject H_0

Not statistical significant

$p\text{-value} > \alpha$

$(1 - \alpha)100\%$ CI contains μ_0

$|\text{test statistics}| < |\text{critical point}|$

Observed results likely due to chance

Data consistent with H_0

Under proper conditions (that is, you will need to consider sample sizes, population distributions, etc.) :

The test statistic for testing $H_0 : \mu = \mu_0$ when σ is known :

$$z_{ts} = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}.$$

The test statistic for testing $H_0 : \mu = \mu_0$ when σ is unknown :

$$t_{ts} = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}.$$

If $H_1 : \mu > \mu_0$, then

$$p\text{-value} = P(T \geq t_{ts}) = \text{pt}(t_{ts}, df, \text{lower.tail} = F).$$

If $H_1 : \mu < \mu_0$, then

$$p\text{-value} = P(T \leq t_{ts}) = \text{pt}(t_{ts}, df).$$

If $H_1 : \mu \neq \mu_0$, then

$$p\text{-value} = P(T \geq |t_{ts}|) \\ = 2 * pt(|t_{ts}|, df, lower.tail = F).$$

If H_1 is one-sided, then the critical point is

$$qt(\alpha, df, lower.tail = F).$$

If H_1 is two-sided, then the critical point is

$$qt(\alpha/2, df, lower.tail = F).$$

Type I error is Falsely rejecting a true null hypothesis. Type II error is Falsely failing to reject a false null hypothesis when the alternative hypothesis is true.

To test $H_0 : \mu_X - \mu_Y = 0$:

If two samples are independent with unequal population variances, then the test statistic is

$$t_{ts} = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}},$$

with

$$df = \frac{(\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y})^2}{\frac{(\frac{s_X^2}{n_X})^2}{n_X - 1} + \frac{(\frac{s_Y^2}{n_Y})^2}{n_Y - 1}}.$$

If two samples are independent with equal population variances, then the test statistic is

$$t_{ts} = \frac{\bar{x} - \bar{y}}{s_p \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}},$$

where

$$df = n_X + n_Y - 2 \text{ and} \\ s_p = \sqrt{\frac{(n_X - 1)s_X^2 + (n_Y - 1)s_Y^2}{n_X + n_Y - 2}}.$$

If sampled data are matched pairs, then the test statistic is

$$t_{ts} = \frac{\bar{D}}{\frac{s_d}{\sqrt{n_d}}}.$$

If a set of data has been given, you may use `t.test()` in R to perform a test. Please refer to R documentation for more info.

To test $H_0 : \frac{\sigma_X^2}{\sigma_Y^2} = 1$, the test statistic is

$$F_{ts} = \frac{s_X^2}{s_Y^2}.$$

If $H_1 : \frac{\sigma_X^2}{\sigma_Y^2} > 1$, then

$$p\text{-value} = pf(F_{ts}, n_X - 1, n_Y - 1, lower.tail = F).$$

If $H_1 : \frac{\sigma_X^2}{\sigma_Y^2} < 1$, then

$$p\text{-value} = pf(F_{ts}, n_X - 1, n_Y - 1).$$

If $H_1 : \frac{\sigma_X^2}{\sigma_Y^2} \neq 1$, then

$$p\text{-value} = 2 * pt(F_{ts}, n_X - 1, n_Y - 1, lower.tail = F), \text{ if } F_{ts} > 1;$$

$$p\text{-value} = 2 * pt(F_{ts}, n_X - 1, n_Y - 1), \text{ if } F_{ts} < 1.$$

If you use the F distribution table to find the p-value, then you need to refer to the lecture notes method to form the test statistic.

If a set of data has been given, you may use `var.test()` in R to perform a test. Please refer to R documentation for more info.

Chapter 9

You will always need to check proper condistions so that the following formulae can be used.

One way ANOVA model is

$$X_{ij} = \mu_i + \epsilon_{ij}$$

where $i = 1, \dots, I$ for levels and $j = 1, \dots, J$ for replicates.

To test $H_0 : \mu_1 = \dots = \mu_I$, use the one way ANOVA table as follows :

Source	df	SS	MS	F	p-value
Factor	I-1	SSTr	MSTr=SSTr/(I-1)	F=MSTr/MSE	p=pf(F, I-1, N-I, lower.tail=F)
Error	N-I	SSE	MSE=SSE/(N-I)		
Total	N-1	SSTotal			

where

$$SSTr = \sum_{i=1}^I J_i (\bar{X}_{i.} - \bar{X}_{..})^2$$

$$SSE = \sum_{i=1}^I (J_i - 1)s_i^2$$

$$SSTotal = SSTr + SSE$$

$$DFTotal = DFT_r + DFE$$

The critical value for one way ANOVA F test is

$$qf(\alpha, I - 1, N - I, lower.tail = F).$$

If the factor/treatment effect is significant, compute the CI for each μ_i as

$$\bar{x}_i \pm t_{N-I, \alpha/2} \sqrt{\frac{MSE}{J_i}}.$$

Two way ANOVA model is

$$X_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ijk}$$

where $i = 1, \dots, I$ for levels of factor A, $j = 1, \dots, J$ for levels of factor B and $k = 1, \dots, K$ for replicates.

To test interaction effects and main effects, use the two way ANOVA table as follows :

Source	df	SS	MS	F	p-value
Factor A	$I - 1$	SSA	$\frac{SSA}{df_A}$	$\frac{MSA}{MSE}$	from $F_{df_A, df_{Error}}$
Factor B	$J - 1$	SSB	$\frac{SSB}{df_B}$	$\frac{MSB}{MSE}$	from $F_{df_B, df_{Error}}$
A * B	$(I - 1)(J - 1)$	SSAB	$\frac{SSAB}{df_{AB}}$	$\frac{MSAB}{MSE}$	from $F_{df_{AB}, df_{Error}}$
Error	$IJ(K - 1)$	SSE	$\frac{SSE}{df_E}$		
Total	$IK - 1$	SST			

The estimated effects are

$$\hat{\alpha}_i = \bar{x}_{i..} - \bar{x}...$$

$$\hat{\beta}_j = \bar{x}_{.j.} - \bar{x}...$$

$$\hat{\gamma}_{ij} = \bar{x}_{ij.} - \bar{x}_{i..} - \bar{x}_{.j.} + \bar{x}...$$

The critical value for a two way ANOVA F test is

$$qf(\alpha, df_{numerator}, df_{denominator}, lower.tail = F).$$

If the interaction effect is not significant but a factor main effect is significant, compute the CI for the factor means of the significant factor(s) as

$$\bar{x}_{i..} \pm t_{IJ(K-1), \alpha/2} \sqrt{\frac{MSE}{JK}} \text{ and/or}$$

$$\bar{x}_{.j.} \pm t_{IJ(K-1), \alpha/2} \sqrt{\frac{MSE}{IK}}.$$

Block design and 2^3 factorial experiments use similar techniques as shown in one-way or two-way factor analysis. Please study these two sections for more information.

If data are given, you may use `lm()`, `anova()` and `confint()`, etc. in R to perform ANOVA tests. Please refer to R documentation for more info.

Chapter 7

You will always need to check proper conditions so that the following formulae can be used.

Correlation coefficient is

$$r = \frac{1}{n-1} \left(\frac{x_i - \bar{x}}{s_X} \right) \left(\frac{y_i - \bar{y}}{s_Y} \right) = cor(x, y).$$

The fitted regression line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x,$$

where

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = r \frac{s_y}{s_x}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The coefficient of determination is

$$r^2 = \frac{SSR}{SSTotal} = \frac{\sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

The residual is

$$e_i = y_i - \hat{y}_i.$$

The standard deviation of errors is

$$s = \sqrt{\frac{(1-r)^2 \sum_{i=1}^n (y_i - \bar{y})^2}{n-2}}$$

The standard deviation of $\hat{\beta}_0$ is

$$s_{\hat{\beta}_0} = s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

The standard deviation of $\hat{\beta}_1$ is

$$s_{\hat{\beta}_1} = \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

Inference about $\hat{\beta}_1$ can be obtained through

$$\hat{\beta}_1 \pm t_{n-2, \alpha/2} s_{\hat{\beta}_1}$$

$$t_{ts} = \frac{\hat{\beta}_1 - \beta_1^{claim}}{s_{\hat{\beta}_1}}$$

Inference about $\hat{\beta}_0$ can be obtained through

$$\hat{\beta}_0 \pm t_{n-2, \alpha/2} s_{\hat{\beta}_0}$$

$$t_{ts} = \frac{\hat{\beta}_0 - \beta_0^{claim}}{s_{\hat{\beta}_0}}$$

Inference about the mean response y for a given x is

$$\hat{y} \pm t_{n-2, \alpha/2} s_{\hat{y}}$$

$$t_{ts} = \frac{\hat{y} - \mu_0}{s_{\hat{y}}}$$

where

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$s_{\hat{y}} = s \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

and μ_0 is the claimed mean response value for the given x value.

Prediction interval for a future y at a given x is

$$\hat{y} \pm t_{n-2, \alpha/2} s_{pred}$$

where

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$s_{pred} = s \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}.$$